NOTICE 1
23 October 2012

TEST OPERATIONS PROCEDURE

This Test Operations Procedure (TOP) has been archived per ATEC memorandum dated 18 October 2012.

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Provides procedures for use of high-speed flash radiographic equipment to obtain shadowgraphs or radiographs of projectile performance in-bore, at the muzzle, in flight, or upon impact with target. Describes equipment, test setup, safety precautions, advantages and limitations.

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U. S. ARMY TEST AND EVALUATION COMMAND
TEST OPERATIONS PROCEDURE

DRSTE-RP-702-103
*Test Operations Procedure 4-2-825 8 June 1978
AD No. A057390

FLASH RADIOGRAPHY IN BALLISTIC TESTING

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1. SCOPE. This TOP provides procedures for using high-speed flash radiographic equipment to obtain shadowgraphs or radiographs showing projectile performance in-bore, at the muzzle, in flight, or when impacting upon a target.

2. FACILITIES AND INSTRUMENTATION.

2.1 Facilities.

ITEM	REQUIREMENTS
Flash X-ray unit (fig. 1 and app. A).	Refer to paragraph 3.1.
Trigger circuit	To provide a triggering signal for pulsing the X-ray unit at the appropriate time (para 3.2.4).
Intensifying screens	Cardboard or plastic support material coated with a chemical such as calcium tungstate which fluoresces immediately after absorbing X-rays and increases the ability of the X-rays to produce usable information on film.

*This TOP supersedes MTP 4-2-825, 25 November 1970.

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ITEM (Cont)	REQUIREMENTS (Cont)
Cassettes (metal holders for the cardboard film holders)	As needed to protect the film and intensifying screens against blast.
Operational shields	Standard and specially modified bomb-proofs to protect equipment from blast and fragmentation.

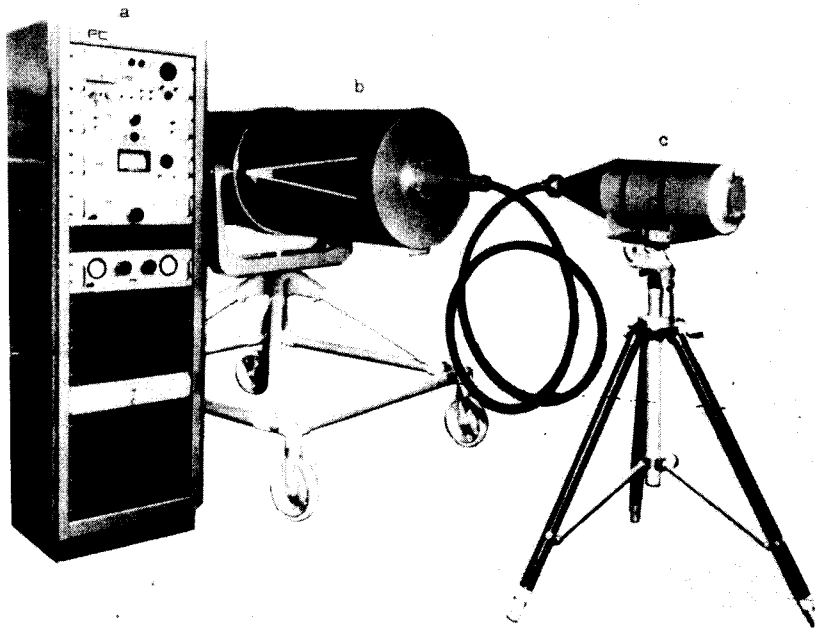


Figure 1. X-Ray Equipment for One-Channel, 600-kV Flash Radiographs, Including (a) Control Panel (High-Voltage Power Supply, Delay Trigger Amplifier, Freon and Nitrogen Gas Regulators), (b) Pulse, and (c) Remote Tubehead.

2.2 Instrumentation.

ITEM	MAXIMUM ERROR OF MEASUREMENT*
Dosimeters (direct reading pocket type)	±10% of full-scale reading for 0-200 mrad range
Film badges	
Pulse event timer	+1 count, -0 count (least significant digit)

*Values may be assumed to represent ±2 standard deviations; thus the stated tolerances should not be exceeded in more than 1 measurement out of 20.

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3. PREPARATION FOR RADIOGRAPHY.

3.1 Planning.

a. Review the ballistic data requirements and the characteristics of the item to be tested. Such information as shape of projectile, location of exterior components, configuration of interior components, damage to target, or simply a general view of a projectile at a particular location may be required.

b. Select a flash-X-ray unit of the proper voltage to obtain the required ballistic data based on the material mass of the test object and the source-to-object distance necessary to avoid potential damage to X-ray equipment. Typical examples of units currently being employed are:

(1) 150 and 180 kV - Radiographs or shadowgraphs of fragment mass and position in lethality studies.

(2) 300 and 600 kV - Radiographs of test objects when penetration of the object is necessary to assess ballistic performance or a large source-to-object distance is necessary to prevent equipment damage.

(3) 2.3 MeV - Radiographs of projectile in-bore or of test objects when penetration of the object is necessary to assess ballistic performance and for recording large-scale ballistic events.

Multichannel units are required when more than a single flash radiograph of a ballistic event is needed. Examples of this requirement are the need for views at two or three distances in front of a gun muzzle, or two sets of orthogonally paired views of fragments coming off the rear of a target plate, for determination of fragment mass and position in lethality studies. A single-channel unit can be built up to a multi-channel capability by adding additional pulsers, delay trigger amplifiers, and X-ray tubes (plus cables and tubeheads, where remote operation is required).

c. Determine the optimum operating parameters for obtaining the desired X-ray penetration (e.g. charging voltage, tube-to-object and object-to-film distances, etc.) from records of past radiographic examinations or by making static radiographs of the test item.

d. Establish test site locations for the X-ray equipment to produce the required radiograph or shadowgraph. The setup will depend to a great extent upon:

(1) Number of views necessary and spatial arrangement of views.

(2) Expected velocity and direction of the test object.

(3) Damage potential to the X-ray equipment.

e. Prepare a test operations checklist using appendix B as a guide and adding specifics for the test item and suitation.

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f. Design data collection sheets for the particular radiographic examination covering such data requirements as listed in paragraphs 3.3 and 5.2.

3.2 Equipment Setup.

3.2.1 X-ray Equipment.

a. Locate the X-ray equipment at the firing site in accordance with requirements determined during planning (para 3.1). Use standard bombproofs or other arrangements of armor plate to protect high-voltage, pulse-forming equipment and controls from blast or fragmentation as appropriate. Figure 2 shows a typical setup. A special armor-plated X-ray trailer may also be used.



Figure 2. Setup for Flash Radiography at Muzzle of 155-mm Gun, Showing Armor Protection and Equipment Arrangement: (a) Armor Plate (Protects Pulser), (b) Small Bombproof in Which Two Tubeheads Are Mounted (Polyethylene Window is in Front of Anodes), and (c) Metal Stand for Cassette.

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b. Install the pulser or remote tubehead containing the X-ray tube in a specially modified bombproof (b, fig. 2) with a slit opening, covered by a sheet of one-half-inch-thick linear high density polyethylene, in front of the X-ray tube anode. Under severe test conditions, the coaxial cable that connects the pulser to the remote tubehead may also have to be protected by armor plate. In multichannel setups, lead shielding is often required between adjacent X-ray tubes to prevent "flashover" on the X-ray films.

3.2.2 Film and Screens. Use of high-speed, screen-type, double-emulsion film is recommended for flash radiography.

a. Place a fluorescent chemical intensifying screen in contact with each side of the film and inclose the film and screens in a regular cardboard X-ray film holder. The intense, short pulses of X-ray energy have little effect on the film unless the intensifying screens are used. Fluorescence of the chemical on these screens produces the radiograph on the film.

b. When protection against blast is required, insert the film holder into a metal cassette between sheets of felt which provide pressure against the film and screens and serve as a shock absorber when the ballistic event occurs.

c. Position the film holder or cassette behind the test object (or its expected location) and directly in front of the X-ray tube(s) using a suitable stand or structure capable of withstanding the shock of the blast when appropriate. The film should be placed as near as possible to the line of trajectory of the test object for maximum clarity of the radiographic image but far enough away to avoid possible damage from blast and fragmentation. When using film holders, provide some means to maintain the film in contact with the intensifying screens such as placing the film holder between sheets of felt and plywood. When metal cassettes are used, do not fasten the cassettes firmly in place. Permitting the blast from the explosion to move the cassettes increases their ability to resist blast damage.

d. Place lead identification numbers or letters and lead reference markers to denote muzzle or target plate position with respect to the film on the front of the film holder to cassette prior to X-ray exposure. NOTE: Data may be lost if this procedure is not followed.

3.2.3 Trigger Circuit. Select and install a trigger circuit to suit the particular ballistic study (i.e., projectile performance in-bore, at the muzzle, in flight, or after impact). Triggering is usually accomplished from a single trigger, with successive pulse intervals controlled by delay timer settings for additional channels. Separate triggers may be used for additional channels; if this is the case, the X-ray units may be operated as entirely separate units. Examples of several types of triggers and their application follow. Some are illustrated in figure 3.

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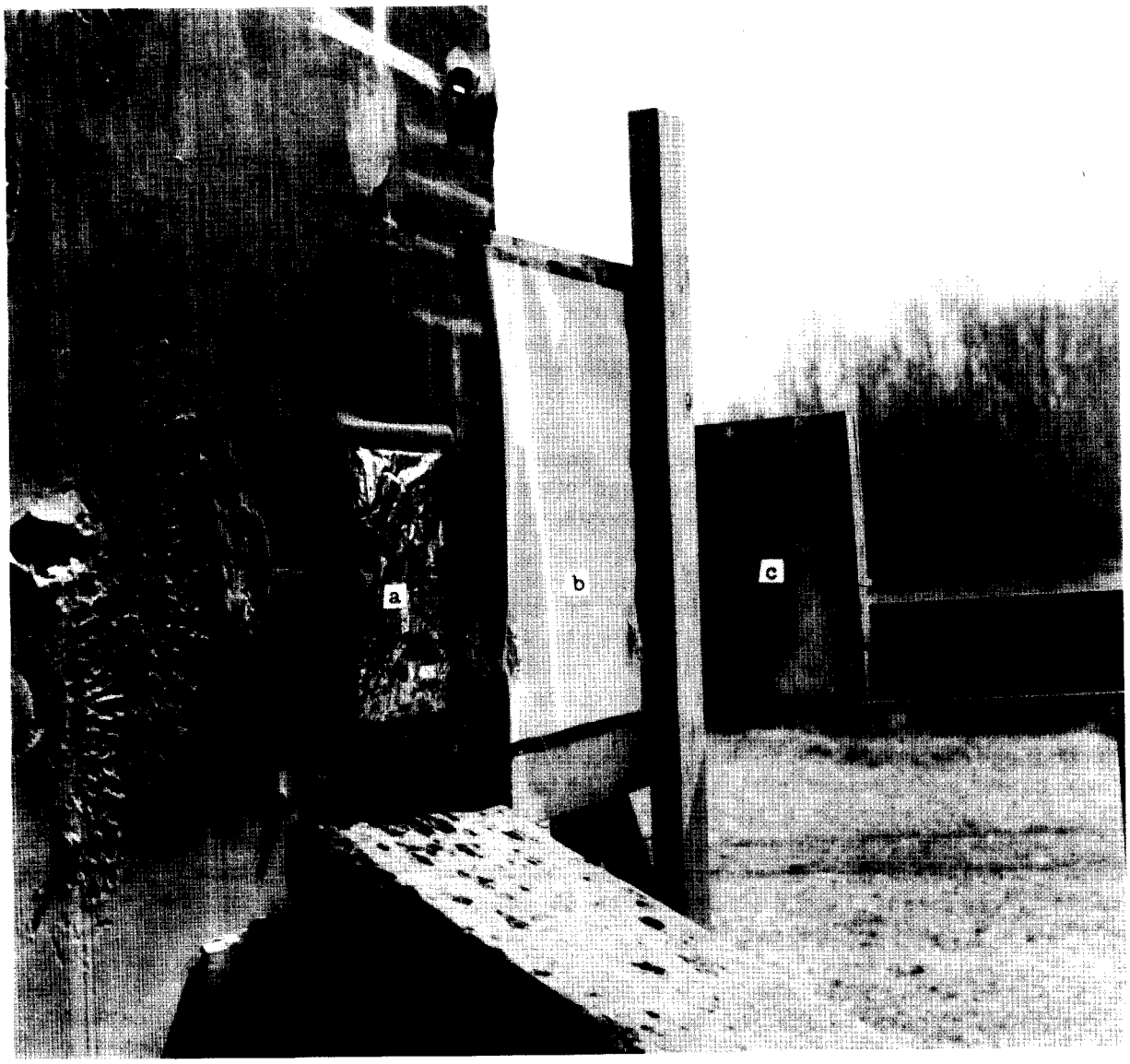


Figure 3. Target Plate Showing (a) Foil "Make" Trigger on Plate, (b) Paper Printed Circuit "Break" Trigger in Front of Plate, and (c) Metal Cassette at Edge of Plate.

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a. Strain gage: Used in an electronic circuit (app. C, fig. 4), usually for studying projectile performance in-bore, where the expansion of the gun tube during projectile travel acts upon the gage (attached to the tube) to produce the pulse.

b. Break screen: Used in an electronic "break" circuit (app. C, fig. 5), usually for terminal ballistic studies, where a projectile or fragment breaks a conductive paint line on the paper screen and causes a pulse to be produced.

c. Blast pressure switch: Used in a "make" circuit, usually at the side of the gun muzzle, where expending gases rebounding off the base of an exiting projectile press a metal diaphragm against a contact.

d. Photocell pickup: Used in a "make" circuit, usually at a gun muzzle, where increasing luminosity produced by the escape of propellant gases is detected.

e. Foil "make" trigger: Aluminum foil backed by cardboard or plastic which makes a circuit when the metal test object shorts the foil to a metal target plate.

3.2.4 Timer Delay Setting. Set the time delay to pulse the X-ray unit as the test object reaches the desired point of travel in front of the film. When an object such as a projectile is moving through air, the time of arrival at the film after triggering can be calculated from standard charge-velocity data. When a projectile or fragment is coming off the back of a target plate, such a calculation cannot usually be made. A time must be estimated and then checked by trial and error. It is often useful to place a thin reference wire in the expected path of the test object, to show whether the timing is fast or slow by the presence or absence of the wire on the X-ray film.

3.3 Data Required.

3.3.1 Test Item.

a. Weapon identification.

b. Projectile identification.

c. Velocity of projectile.

d. Modifications to projectile to enhance radiographic image.

e. Target (if applicable) description.

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3.3.2 Equipment Setup.

a. Number of channels.

b. Time delay between channels.

c. Film size, type, and containment.

d. Distance, film to object.

e. Distance, X-ray tube(s) to object.

f. Distance of film downrange from muzzle or uprange from target.

g. Type and location of triggering device.

h. Protection to X-ray tube(s).

i. Protection for pulser.

j. X-ray unit used.

k. Type of intensifying screens.

l. Photograph of setup.

4. RADIOGRAPHY CONTROLS.

a. Film badges are worn by all personnel assigned to X-ray operations.

b. Each operator carries at least one dosimeter, preferably two. Dosimeter readings are recorded daily.

c. Exposure areas are posted with radiation warning signs in accordance with AR 385-30.*

d. All range and facility safety SOP's are observed throughout the radiographic examination.

e. Qualified X-ray personnel operate the flash X-ray equipment in accordance with the equipment manuals.

5. RADIOGRAPHIC EXAMINATION.

5.1 Method.

a. Prepare the X-ray unit for operation in accordance with the operating manual.

*AR 385-30, Safety Color Code Markings and Signs.

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b. Check whether the expected X-ray dose is being produced in front of the tube. Direct-reading pocket dosimeters are useful for this purpose. The actual dose at any distance can be calculated from the inverse square law,

$$\frac{I_1}{I_2} = \frac{d_2^2}{d_1^2}$$

where I_1 is the measured or known dose at a given distance, d_1, and I_2 is the dose at another distance, d_2.

c. Fire the weapon to function the X-ray triggering circuit according to the particular firing program.

5.2 Data Required.

a. Date and time of examination.

b. Dosimeter readings.

c. Film number versus round number for each firing.

6. DATA REDUCTION AND PRESENTATION.

a. Check that the X-ray film holders contain identification numbers or letters (para 3.2.2d); if they do not, mark the films using a film perforator as soon as they are brought into the darkroom, before processing.

b. Process the film in accordance with appropriate film developing procedures. Use freshly prepared processing chemicals to obtain maximum penetration.

c. Read the radiograph and record the factors appropriate to the specific test (app. A, para 3). Most information is obtainable by visual inspection. Measurements of projectile attitude or shape may be made by comparison of the image with the lead reference markings. High-intensity X-ray film viewers are useful in detecting fine details in films with high photographic densities.

d. When very accurate film measurements are necessary, allow for image magnification created by the geometry of the setup. Known dimensions of the object usually can be measured on the film to provide a magnification ratio for correcting other measured distances on the film. (Reference markers attached to the cassette, being closer to the film than the test object, cannot be used to scale measurements on the object.)

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APPENDIX A
FLASH RADIOGRAPHIC EQUIPMENT

1. Description. Radiography is one of the widely used nondestructive test techniques. Flash radiography is a special adaptation which was first used in the United States for studies of shaped-charge jets. It differs from conventional, industrial radiography in that the X-ray energy is delivered in the form of a pulse of fractional microsecond duration as opposed to the continuous-level output of industrial machines.

Basic flash radiographic equipment (fig. 1, p. 2) consists of a Marx-type generator, charging transformer, triggering circuit, and a cold-cathode-field emission X-ray tube. The Marx generator, or pulser, is made up of a bank of capacitors arranged so that they are charged in parallel then switched through sphere gaps to discharge in series. A pressurized dry nitrogen system insures a constant operating atmosphere for the sphere gap discharge. In operation, a trigger pulse is amplified by a transformer to an amplitude sufficient to cause a voltage breakdown across the initial sphere gap. The series discharge follows and is impressed across the X-ray tube.

2. Advantages and Limitations. During ballistic testing it is often necessary to obtain projectile performance data that cannot be obtained by means of high-speed cameras. With camera coverage the location and shape of projectile parts in-bore, at the muzzle, and upon impact with a target usually are obscured by flash, smoke, debris, and other intervening materials. Flash radiographic equipment, on the other hand, can provide views of the more dense projectile parts with little interference from the less dense intervening materials. In this sense, flash radiography may be used to supplement regular photographic coverage. There is the additional advantage, however, of the view's being a radiograph as well as an outline or shadowgraph of the test object.

Flash radiography has been used in studies of shaped charges; armor-piercing, discarding-sabot projectiles; defeat of special armor arrangements; and interaction of explosive projectiles with targets, among other applications.

Most flash radiographic equipment is limited by relatively low radiation output to a fairly small amount of true radiographic utility where internal parts of items can be viewed. Specially constructed projectile components may be necessary, on occasion, to permit the viewing of internal parts.

Another limitation of this technique is the fact that X-rays cannot be focused in a practical sense, thus images must be recorded full scale on large sheets of film. This prevents a rapid movement of film (such as in a movie camera) and limits the number of views or frames in sequence of one dynamic event.

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3. Information Generated.

a. Dynamic Behavior of In-Bore/Out-of-Bore Projectile or Fragment.

Parameter	Uncertainty of Data at One Standard Deviation
Yaw pitch	±0.1°
Muzzle, striking, and between plate velocity	±3 m/s (±10 fps)
Exiting plate velocity	±7.6 m/s (±25 fps)
Mass	±10%
Kinetic energy	±10%
Dispersion	±10%
Probability of Incapacitation	±10%
Crush-up	±0.25 cm (±0.1 in.)

b. Dynamic Behavior of a Shaped Charge.

Parameter	Uncertainty of Data at One Standard Deviation
Jet length	±0.5%
Jet width	±0.5%
Particle velocity	±1.0%
Particle mass	±10%
Kinetic energy of particle	±10%
Position	±0.25 cm (±0.1 in.)

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APPENDIX B
CHECKLIST GUIDE FOR RADIOGRAPHIC EXAMINATION OF BALLISTICS

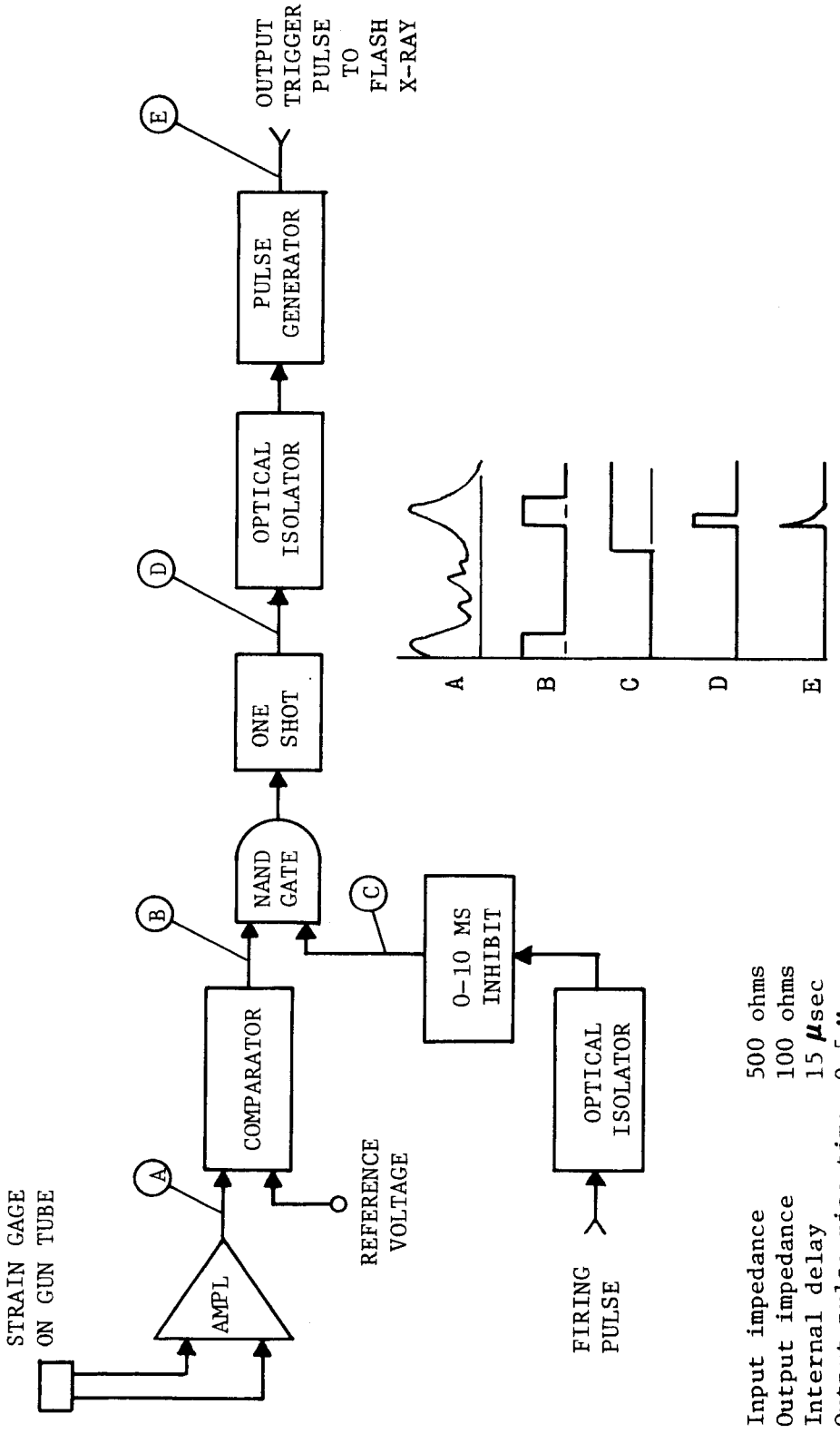
ITEM	YES	NO	NA
1. All operating personnel briefed on test requirements, special procedures, hazards, and any unusual aspects of test.			
2. Test item and equipment setup data recorded (para 3.3).			
3. Instrumentation calibrated, properly installed, and operational.			
4. X-ray tube protected.			
5. X-ray dosage checked.			
6. Lead identification numbers or letters and lead reference marks positioned in front of film holder or cassette.			
7. Film badges worn by personnel.			
8. Exposure areas posted in accordance with AR 385-30.			
9. Safety requirements accomplished and SOPs posted at test site.			

*SOPs 385-12 and 67 at Aberdeen Proving Ground

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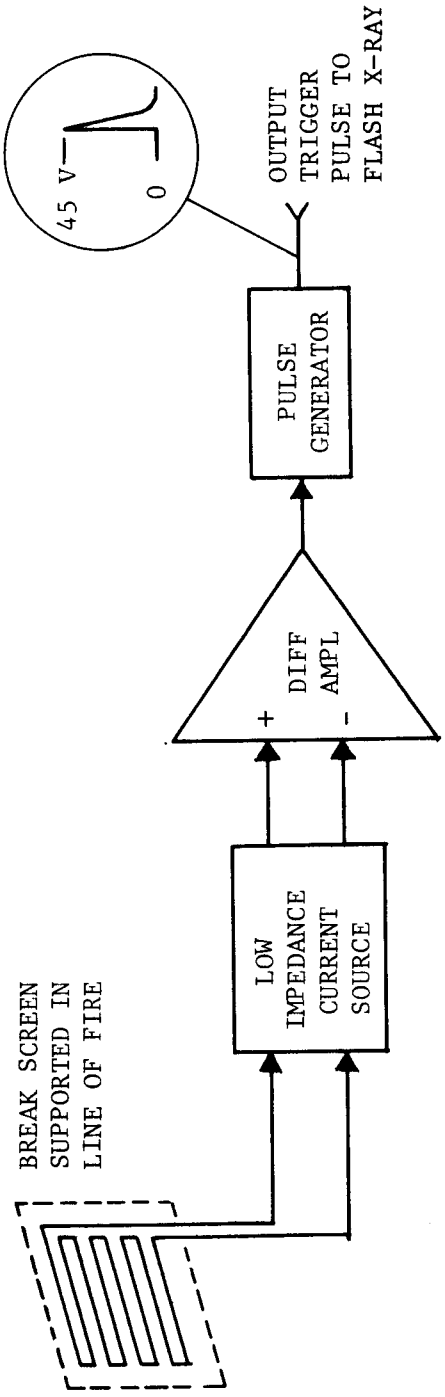
APPENDIX C
TRIGGER CIRCUITS



Input impedance	500 ohms
Output impedance	100 ohms
Internal delay	15 μsec
Output pulse rise time	0.5 μsec
Output pulse voltage	45 V
Sensitivity	0.2 ohm

Figure 4. Strain Gage Trigger (Battery Powered).

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Input impedance	100 ohms
Output impedance	100 ohms
Internal delay	<2 μsec
Output pulse rise time	<0.5 μsec
Output pulse voltage	45 V
Sensitivity	0 to 1000 ohms

Figure 5. Break Screen Trigger (AC Powered).

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